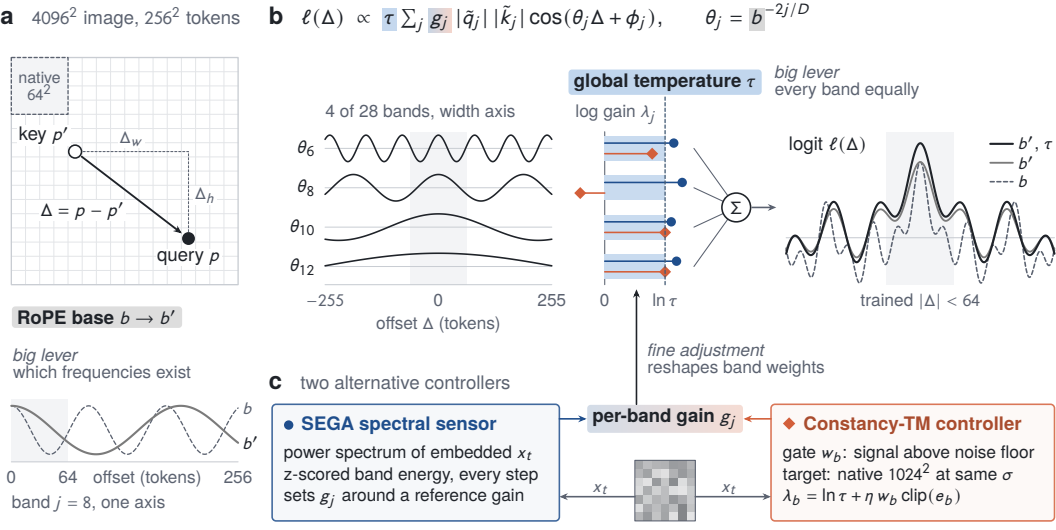
a
4096² image, 256² tokens
native 64²
key p′
Δw
Δh
Δ = p − p′
query p
RoPE base b → b′
big lever
which frequencies exist
b
b′
0
64
offset (tokens)
256
band j = 8, one axis
b
ℓ(Δ) ∝ τ Σj gj |q̃j| |k̃j| cos(θj Δ + φj),
θj = b^−2j/D
4 of 28 bands, width axis
θ6
θ8
θ10
θ12
−255
0
255
offset Δ (tokens)
global temperature τ
log gain λj
0
ln τ
Σ
big lever
every band equally
logit ℓ(Δ)
b′, τ
b′
b
trained |Δ| < 64
fine adjustment
reshapes band weights
c
two alternative controllers
SEGA spectral sensor
power spectrum of embedded xt
z-scored band energy, every step
sets gj around a reference gain
per-band gain gj
xt
xt
Constancy-TM controller
gate wb: signal above noise floor
target: native 1024² at same σ
λb = ln τ + η wb clip(eb)